

1.3.1	Montelukast sodium
SPC, Labeling and Package Leaflet	MY

NAME OF THE MEDICINAL PRODUCT

Monkasta film-coated tablets 10 mg

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each film-coated tablet contains 10 mg montelukast (equivalent to 10.4 mg montelukast sodium).

Excipient:

	10 mg tablets
lactose	81.94 mg

PHARMACEUTICAL FORM

Film-coated tablet.

The film-coated tablets are apricot-coloured, round, slightly biconvex, with bevelled edges.

CLINICAL PARTICULARS

Therapeutic indications

For the prophylaxis and chronic treatment of asthma in adults and paediatric patients 12 months of age and older.

Monkasta is indicated if adults 15 years of age and older for the relief of daytime and night-time symptoms of seasonal allergic rhinitis.

Posology and method of administration

Monkasta should be taken once daily. For the treatment of asthma, the dose should be taken in the evening. For seasonal allergic rhinitis, the time of administration may be individualised to suit patient needs.

Adults 15 years of Age and Older with Asthma and/or Seasonal Allergic Rhinitis

The dosage for adults 15 years of age and older is one 10 mg tablet daily.

General recommendations

The therapeutic effect of Monkasta on parameters of asthma control occurs within one day.

Monkasta may be taken with or without food.

Patients should be advised to continue taking Monkasta even if their asthma is under control, as well as during periods of worsening asthma.

Monkasta should not be used concomitantly with other products containing the same active ingredient, montelukast.

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No dosage adjustment is necessary for the elderly, or for patients with renal insufficiency, or mild to moderate hepatic impairment. There are no data on patients with severe hepatic impairment. The dosage is the same for both male and female patients.

Therapy with Monkasta in relation to other treatments for asthma

Monkasta can be added to a patient's existing treatment regimen.

β-agonist therapy: Monkasta can be added to the treatment regimen of patients who are not adequately controlled on 'as-needed' short-acting β-agonist. When a clinical response is evident (usually after the first dose), the patient may be able to decrease the use of 'as-needed' short-acting β-agonist.

Inhaled corticosteroids: Treatment with Monkasta can be used as add-on therapy in patients when other agents, such as inhaled corticosteroids, provide inadequate clinical control. Monkasta should not be substituted for inhaled corticosteroids (*See Section 4.4*).

Method of administration

Oral use

Contraindications

Hypersensitivity to the active substance or to any of the excipients.

Special warnings and precautions for use

Patients should be advised never to use oral montelukast to treat acute asthma attacks and to keep their usual appropriate rescue medication for this purpose readily available. If an acute attack occurs, a short-acting inhaled β-agonist should be used. Patients should seek their doctor's advice as soon as possible if they need more inhalations of short-acting β-agonists than usual.

Montelukast should not be substituted for inhaled or oral corticosteroids.

There are no data demonstrating that oral corticosteroids can be reduced when montelukast is given concomitantly.

In rare cases, patients on therapy with anti-asthma agents including montelukast may present with systemic eosinophilia, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition which is often treated with systemic corticosteroid therapy. These cases usually, but not always, have been associated with the reduction or withdrawal of oral corticosteroid therapy. The possibility that leukotriene receptor antagonists may be associated with emergence of Churg-Strauss syndrome can neither be excluded nor established. Physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients. Patients who develop these symptoms should be reassessed and their treatment regimens evaluated.

Treatment with montelukast does not alter the need for patients with aspirin-sensitive asthma to avoid taking aspirin and other non-steroidal anti-inflammatory drugs.

Special information about some of the ingredients

Monkasta contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this drug.

Interaction with other medicinal products and other forms of interaction

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Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma. In drug-interactions studies, the recommended clinical dose of montelukast did not have clinically important effects on the pharmacokinetics of the following drugs: theophylline, prednisone, prednisolone, oral contraceptives (ethinyl oestradiol/norethindrone 35/1), terfenadine, digoxin and warfarin.

The area under the plasma concentration curve (AUC) for montelukast was decreased approximately 40% in subjects with co-administration of phenobarbital. Since montelukast is metabolised by CYP 3A4, caution should be exercised, particularly in children, when montelukast is co-administered with inducers of CYP 3A4, such as phenytoin, phenobarbital and rifampicin.

In vitro studies have shown that montelukast is a potent inhibitor of CYP 2C8. However, data from a clinical drug-drug interaction study involving montelukast and rosiglitazone (a probe substrate representative of drugs primarily metabolised by CYP 2C8) demonstrated that montelukast does not inhibit CYP 2C8 *in vivo*. Therefore, montelukast is not anticipated to markedly alter the metabolism of drugs metabolised by this enzyme (e.g., paclitaxel, rosiglitazone, and repaglinide).

Pregnancy and lactation

Use During Pregnancy

Limited data from available pregnancy databases do not suggest a causal relationship between montelukast and malformations (i.e. limb defects) that have been rarely reported in worldwide post marketing experience.

Montelukast may be used during pregnancy only if it is considered to be clearly essential.

Use During Lactation

It is not known if montelukast is excreted in human milk. Montelukast may be used in nursing mothers only if it is considered to be clearly essential.

Effects on ability to drive and use machines

There is no evidence that montelukast affects the ability to drive and use machines. However, in very rare cases, individuals have reported drowsiness.

Undesirable effects

Montelukast has been generally well tolerated. Side effects, which were usually mild, generally did not require discontinuation of therapy. The incidence of side effects reported with montelukast was comparable to placebo.

The following drug-related adverse reactions in placebo-controlled clinical studies were reported commonly in asthmatic patients treated with montelukast and at a greater incidence than in patients treated with placebo.

Post marketing Experience

The following side reactions have been reported in post marketing use: hypersensitivity reactions (including anaphylaxis, angioedema, rash, urticaria and very rarely hepatic eosinophilic infiltration); dream abnormalities and hallucinations, drowsiness, dizziness, irritability, agitation including aggressive behaviour, obsessive-compulsive symptoms, restlessness, insomnia, paraesthesia/hypoesthesia, and very rare seizures; nausea, vomiting, dyspepsia, diarrhoea, increased ALT and AST, and very rarely cholestatic hepatitis, arthralgia, myalgia including muscle cramps; thrombocytopenia, increased bleeding tendency, bruising; palpitations; and oedema.

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Very rare cases of Churg-Strauss Syndrome (CSS) have been reported during montelukast treatment in asthmatic patients. (see Section 4.4).

Overdose

No specific information is available on the treatment of overdosage with montelukast. Both short-term and long-term exposure to daily doses significantly exceeding the recommended dose (200 – 900 mg/day) did not result in clinically significant adverse events.

There were no adverse experiences in the majority of overdosage reports. The most frequently occurring adverse experiences were consistent with the safety profile of montelukast and included abdominal pain, somnolence, thirst, headache, vomiting, and psychomotor hyperactivity.

It is not known whether montelukast is dialysable by peritoneal- or haemo-dialysis.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: Anti-Asthmatics for systemic use, Leukotriene receptor antagonist. ATC Code: R03DC03.

The cysteinyl leukotrienes (LTC₄, LTD₄, LTE₄) are potent inflammatory eicosanoids released from various cells including mast cells and eosinophils. These important pro-asthmatic mediators bind to cysteinyl leukotriene (CysLT) receptors. The CysLT type-1 (CysLT₁) receptor is found in the human airway (including airway smooth muscle cells and airway macrophages) and on other proinflammatory cells (including eosinophils and certain myeloid stem cells). CysLTs have been correlated with the pathophysiology of asthma and allergic rhinitis. In asthma, leukotriene-mediated effects include bronchoconstriction, mucous secretion, vascular permeability, and eosinophil recruitment. In allergic rhinitis, CysLTs are released from the nasal mucosa after allergen exposure during both early- and late-phase reactions and are associated with symptoms of allergic rhinitis. Intranasal challenge with CysLTs has been shown to increase nasal airway resistance and symptoms of nasal obstruction.

Montelukast is an orally active compound which binds with high affinity and selectivity to the CysLT₁ receptor. Montelukast inhibits bronchoconstriction due to inhaled LTD₄ at doses as low as 5 mg. Bronchodilation was observed within two hours of oral administration. The bronchodilation effect caused by a β -agonist was additive to that caused by montelukast. Treatment with montelukast inhibited both early- and late-phase bronchoconstriction due to antigen challenge.

Pharmacokinetic properties

Absorption:

Montelukast is rapidly absorbed following oral administration. For the 10 mg film-coated tablet, the mean peak plasma concentration (C_{max}) is achieved three hours (T_{max}) after administration in adults in the fasted state. The mean oral bioavailability is 64%. The oral bioavailability and C_{max} are not influenced by a standard meal. Safety and efficacy were demonstrated in clinical trials where the 10 mg film-coated tablet was administered without regard to the timing of food ingestion.

Distribution:

Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of montelukast averages 8-11 litres. Studies in rats with radiolabelled montelukast indicate minimal

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distribution across the blood-brain barrier. In addition, concentrations of radiolabelled material at 24 hours post-dose were minimal in all other tissues.

Biotransformation:

Montelukast is extensively metabolised. In studies with therapeutic doses, plasma concentrations of metabolites of montelukast are undetectable at steady state in adults and children.

In vitro studies using human liver microsomes indicate that cytochromes P450 3A4, 2A6 and 2C9 are involved in the metabolism of montelukast. Based on further *in vitro* results in human liver microsomes, therapeutic plasma concentrations of montelukast do not inhibit cytochromes P450 3A4, 2C9, 1A2, 2A6, 2C19, or 2D6. The contribution of metabolites to the therapeutic effect of montelukast is minimal.

Elimination:

The plasma clearance of montelukast averages 45 ml/min in healthy adults. Following an oral dose of radiolabelled montelukast, 86% of the radioactivity was recovered in 5-day faecal collections and <0.2% was recovered in urine. Coupled with estimates of montelukast oral bioavailability, this indicates that montelukast and its metabolites are excreted almost exclusively *via* the bile.

Characteristics in patients:

No dosage adjustment is necessary for the elderly or mild to moderate hepatic insufficiency. Studies in patients with renal impairment have not been undertaken. Because montelukast and its metabolites are eliminated by the biliary route, no dose adjustment is anticipated to be necessary in patients with renal impairment. There are no data on the pharmacokinetics of montelukast in patients with severe hepatic insufficiency (Child-Pugh score>9).

With high doses of montelukast (20- and 60-fold the recommended adult dose), a decrease in plasma theophylline concentration was observed. This effect was not seen at the recommended dose of 10 mg once daily.

PHARMACEUTICAL PARTICULARS

Shelf life

3 years.

Special precautions for storage

Do not store above 30°C.
Store in the original package in order to protect from light.

Nature and contents of container

Blister pack of 28 film-coated tablets in a box.

PRODUCT REGISTRATION HOLDER

PAHANG PHARMACY SDN. BHD.
Lot 5979, Jalan Teratai, 5 ½ Miles, Off Jalan Meru, 41050 Klang, Selangor, Malaysia

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DATE OF REVISION OF THE TEXT

April 2021